

High-Level Structure of Dependency Networks

Alexei Vazquez

Nodes & Links Ltd, Salisbury House, Station Road, Cambridge, CB1 2LA, UK

alexei@nodeslinks.com

Abstract

Dependency networks describe how activities, events or processes are linked by prerequisite relationships, but real networks are often too large to interpret directly. Here we ask how to summarise such a network at a higher level of organisation, using construction project schedules as a case study. We introduce two simple, deterministic ways of grouping activities: the Dependency Uprise Structure (DUS), which collects activities that share the same logical depth in the network, and the Dependency ~~Fibration Structure (DFS)~~ Predecessor Structure (DPS), which refines this grouping by also recording which earlier levels each activity depends on. ~~Applied to construction projects, the DUS compresses the original~~ Across 1198 construction project schedules from three companies, spanning 36 to 150,692 dependencies, the number of DUS levels grows only logarithmically with the number of dependencies, while the number of DPS classes grows as a sub-linear power law with exponent 0.71 (95% CI [0.69, 0.72]). Model selection favours these two functional forms over the alternatives we considered, and the logarithmic coefficient and the power-law exponent are statistically indistinguishable across the three companies. The DUS therefore compresses the dependency network exponentially and the ~~DFS~~ DPS provides an intermediate, sub-linear summary. ~~Both reveal a small-world organisation that is hidden in the raw schedule.~~ The mean distance from a DUS level to the end of the project is almost independent of project size, because dependencies that skip levels act as shortcuts; a degree-preserving null model shows that this shallowness follows from the degree distributions of the DUS network rather than from any special arrangement of those shortcuts. We replicate the qualitative results on the publicly available PSPLIB benchmark. Because dependency networks appear in fields as diverse as project management, causal inference, ancestry analysis and neural networks, the same approach should generalise ~~well~~ beyond construction. A Python implementation is available at <https://github.com/av2atgh/dus>.

Introduction

Most large human endeavours are organised as projects [1, 2, 3]. A building is constructed, a piece of software is delivered, or a new drug is brought through clinical trials by coordinating many smaller activities under shared deadlines and budgets. Although the products are very different, the underlying logistics are remarkably similar, and the discipline of project management has emerged to plan and control them across sectors.

Coordinating a project of any size requires a high-level framework that connects day-to-day execution with strategic goals [4, 5]. Almost universally, this is achieved by breaking the total work into smaller, manageable units that can be assigned to teams, costed, scheduled and tracked. The way the work is broken down determines how progress is monitored and how risks are spotted.

The standard tool for this decomposition is the Work Breakdown Structure (WBS) [4]. The WBS is a tree of deliverables: the final product sits at the top and is recursively divided into ever smaller pieces of work, until the leaves are activities small enough to be scheduled and budgeted individually. Figure 1A shows a deliberately small example for a house: the project divides into substructure, superstructure and services, and these divide in turn into the individual activities

that will appear on the schedule. Real construction WBSs follow exactly this pattern but reach tens of thousands of leaves. The resulting hierarchy is the backbone of project planning, organising responsibilities and providing a common map for everyone involved.

The WBS captures what needs to be done, but not the order in which it has to happen. To turn a list of activities into a schedule, the project manager must specify which activities are prerequisites for which others — the ~~foundation~~ foundations of a building must be poured before the frame goes up, or a back-end service must be available before its front-end can be tested. The set of these prerequisite relationships forms a *dependency network* (Figure 1B). From this network one can compute the critical path, identify slack, and estimate when the project will end [6].

Formally, a dependency network is a directed graph [25] in which each node is an activity and each directed link encodes a prerequisite, with $A \rightarrow B$ meaning that activity A must finish before activity B can start. Such a network must be acyclic: a loop such as $A \rightarrow B \rightarrow C \rightarrow A$ would require A to be finished before it can begin. Dependency networks are therefore directed acyclic graphs (DAGs). Every DAG admits a *topological order*: an arrangement of its nodes in which all of an activity’s predecessors come before it and all of its successors come after it. Some orderings are forced, as in the chain $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_n$, while others leave room for choice. If A_1 feeds both A_2 and A'_2 and both must finish before A_3 starts, then A_2 and A'_2 can be swapped without changing anything that matters for the schedule. Two nodes that can be swapped in this way are said to be *incomparable*: neither is an ancestor of the other. DAGs are the natural representation not only of schedules but also of causal models, ancestries and feedforward neural networks, which is why the constructions below are not specific to construction.

Project teams therefore look at their work through two different lenses: the deliverable-based hierarchy of the WBS and the prerequisite-based wiring of the dependency network. These two views rarely line up, because dependencies routinely cross WBS boundaries — in Figure 1 the utilities activity, which belongs to the services branch, depends on excavation, which belongs to the substructure branch. What is missing is a high-level summary of the dependency network itself — not a top-down decomposition imposed from above, but a structure that emerges from the dependencies. Recovering such a summary is an inverse problem: instead of starting with a goal and dividing it, we start with the fine-grained dependencies and try to expose their large-scale organisation.

~~Network science has long studied related inverse problems, developing increasingly sophisticated statistical methods to find communities and other~~ Several literatures bear on this problem. In project management and engineering design, the Design Structure Matrix (DSM) represents dependencies as a square matrix and has a long tradition of reordering and clustering that matrix to reveal blocks of tightly coupled work and to sequence it [7, 8, 9]. DSM sequencing and the layering used here are close relatives: both seek an ordering compatible with the dependencies. They differ in aim. DSM methods are usually applied to problems that contain cycles, where the object is to find and break feedback loops, and clustering is typically the result of an optimisation with tunable objectives. The constructions below apply to acyclic networks, involve no optimisation and no free parameters, and are aimed at compression rather than resequencing. In computer science, condensing a directed graph by its strongly connected components [10] is the standard way of turning an arbitrary digraph into a DAG; it leaves an already-acyclic network untouched and so cannot compress a schedule. Transitive reduction [11] removes redundant edges but keeps every node. Layered graph drawing [26, 12] assigns nodes to layers by essentially the rule used below, but as a step towards a drawing rather than as an object of study. In network science, meso-scale structure ~~in graphs~~ is usually sought statistically, by inferring communities or blocks [20, 21, 22, 23, 24]; other approaches exploit exact symmetry [13, 14], graph fibrations [30, 31], latent hyperbolic geometry [15], or explicit measures of hierarchy [16, 17]. These methods have rarely been applied to dependency networks. As we show below, the dependency case is ~~in fact~~ considerably simpler: a useful high-

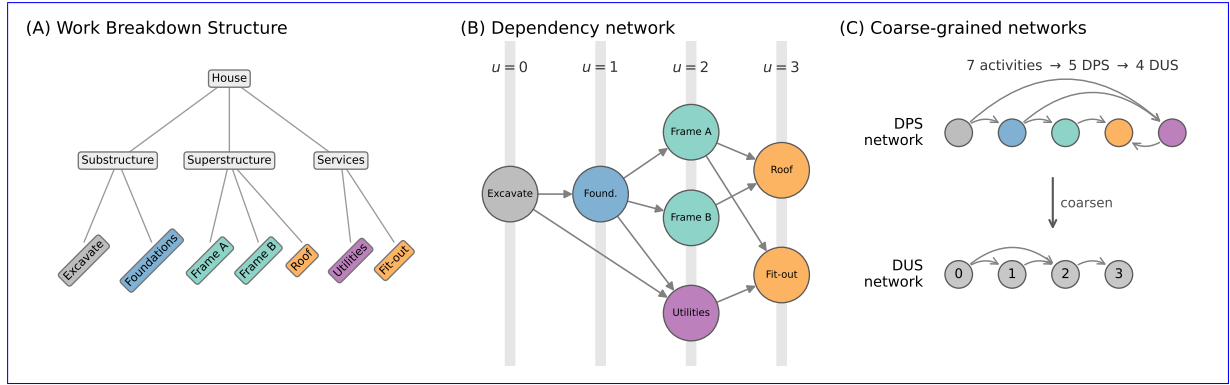


Figure 1: The two coarse-grainings, illustrated on a seven-activity project. (A) A Work Breakdown Structure: the project is recursively divided into deliverables until the leaves are schedulable activities. (B) The dependency network of those activities. The DUS index u of an activity is the length of the longest chain of dependencies leading to it, shown by the vertical bands. Note that the dependency from Excavate to Utilities crosses a WBS branch boundary. Colours indicate the DPS class, that is, the set of DUS levels the activity’s predecessors occupy: Frame A and Frame B share a DUS level and a DPS class, whereas Utilities shares their DUS level but forms a separate DPS class because it also depends on level 0. (C) The two coarse-grained networks obtained by collapsing the dependency network onto DPS classes and onto DUS levels. Seven activities reduce to five DPS classes and four DUS levels.

level structure can be extracted deterministically, without statistical inference and without free parameters, by exploiting the directional, acyclic nature of the network.

The rest of this paper defines the two groupings, illustrated in Figure 1, measures how they scale on 1198 real construction schedules, and asks how far apart the resulting coarse-grained networks are.

Results

Dependency Uprise Structure

~~In a dependency network each node is an activity and each directed link encodes a prerequisite, with $A \rightarrow B$ meaning that activity A must finish before activity B can start. Such networks are directed graphs [25]. They must also be acyclic: a loop such as $A \rightarrow B \rightarrow C \rightarrow A$ would require A to be finished before it can begin. Dependency networks are therefore directed acyclic graphs (DAGs) [25].~~

~~Every DAG admits a topological order: an arrangement of its nodes in which all of an activity’s predecessors come before it and all of its successors come after it. Some orderings are forced, as in the chain $A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow \dots \rightarrow A_n$, while others leave room for choice. If A_1 feeds both A_2 and A'_2 and both must finish before A_3 starts, then A_2 and A'_2 can be swapped without changing anything that matters for the schedule. This freedom motivates assigning We assign every activity an integer that measures how many dependency steps separate it from a starting activity, ~~made precise below.~~~~

~~We assign.~~ We set the DUS index $u_i = 0$ ~~to for~~ every activity with no predecessors, and otherwise

$$u_i = 1 + \max_{j \in \text{PredecessorsOf}(i)} u_j. \quad (1)$$

After processing all activities in a topological order, u_i equals the length of the longest ~~dependency chain~~ chain of dependencies leading to activity i . ~~The~~ We call it the uprise index because it

counts how far the activity has risen above the start of the project, in dependency steps rather than in calendar time.

Three distinct objects follow from Eq. (1), and we keep them separate throughout. The *DUS index* u_i is an integer attached to each activity. A *DUS level* is the set of all activities sharing one value of u_i . The *Dependency Uprise Structure* (DUS) is the partition of activities by this index: each DUS subset contains the activities sharing one value of u_i , resulting partition of the activities into levels. The DUS network, defined in the next section, is a graph whose nodes are DUS levels.

By construction, two activities sharing a DUS index in the same DUS level are pairwise incomparable in the dependency DAG — neither is a predecessor an ancestor of the other — so they can appear in either relative order within a valid topological sort. The converse does not hold: two activities at different depths can also be incomparable, so the DUS is one specific antichain decomposition of the DAG, not in general the coarsest one.

~~The construction in~~ It is worth being explicit about what Eq. (1) is not. It is not a breadth-first search from the project start, which would assign each activity the length of the *shortest* path from a starting activity. The two differ whenever an activity has predecessors at different depths: in Figure 1B, a breadth-first search from Excavate would place Utilities at depth 1, because Excavate feeds it directly, whereas Eq. (1) is analogous to the places it at $u = 2$, because it must also wait for Foundations. Only the longest-path rule has the property that matters here, namely that all of an activity’s predecessors lie strictly below it, so that the levels are ordered and every activity is preceded by the entire history it depends on. It is also not a breadth-first search backwards from the project-completion node, which would group activities by their remaining depth to the end rather than by their accumulated depth from the start.

The construction in Eq. (1) is the layering used in the first phase of the Sugiyama framework for hierarchical graph drawing [26]–[26, 12] and is closely related to the Coffman–Graham scheduling algorithm [27]. ~~The novel~~ In project management terms, the DUS index of the final activity is the number of activities on the longest dependency chain, which is the critical path when all activities have equal duration. The contribution of this paper is not the layering itself but its use as a coarse-graining of the dependency network, the empirical scaling laws established below for construction project schedules relations established below, and the finer DFS construction introduced in the section on the DUS fine structure DPS construction introduced later.

The DUS network

Once activities have been grouped into DUS levels, we can collapse the original dependency network onto these groups. In the resulting DUS network each node is a DUS level, and a weighted edge $U_a \xrightarrow{w_{ab}} U_b$ records that w_{ab} activity-level dependencies run from members of U_a to members of U_b activities in level a to activities in level b . The DUS network inherits the directed acyclic structure of the dependency network. What is new is the weight w_{ab} , which measures how strongly two DUS levels are coupled.

The simplest DUS network is a chain, for example $U_0 \xrightarrow{w_{01}} U_1 \xrightarrow{w_{12}} U_2$. In general, dependencies may also skip ahead, linking distant levels. A chain through every level is, however, always present: by the definition of u_i in Eq. (1), every activity has at least one predecessor in the previous DUS level, so $w_{i-1,i} \geq 1$. The chain $U_0 \rightarrow U_1 \rightarrow U_2 \rightarrow \dots \rightarrow U_{m-1} U_0 \rightarrow U_1 \rightarrow \dots \rightarrow U_{m-1}$, where m is the number of DUSs DUS levels, is therefore the longest path in the DUS network. We call it the *DUS spine*, by analogy to the activity-level critical path while avoiding overlap with that term’s specific duration-weighted meaning in project management.

Figure 1A, B shows the DUS networks of two construction project schedules drawn from the Nodes & Links database (see Methods). Each DUS appears as a dot, arranged from left to right

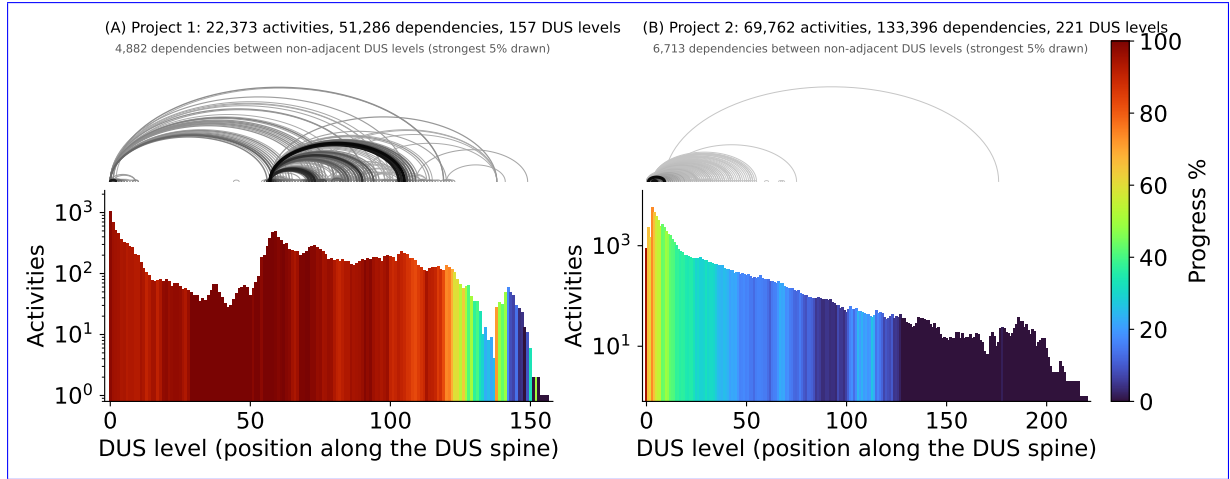


Figure 2: DUS networks. (A,B) DUS networks profiles of two construction project schedules. The nodes represent DUSs, ordered from left to right according to their horizontal axis is position along the DUS spine. The links represent dependencies between DUSs, with Bars give the grey scale proportional to their number. (C, D) Adding the number of activities in each DUS (bar heights) level and current are coloured by the mean Physical Progress Percent of those activities, a standard progress (colour map) metric in construction. Arcs above show dependencies between non-adjacent DUS levels, with darker and thicker arcs encoding more dependencies; only the strongest 5% are drawn, since drawing all of them obscures the structure. Both schedules are included in the public code repository.

Company	Projects	Activities	Dependencies	DUS levels	DPS classes
		median [minimum–maximum]			
Blue	82	43,167 [83–70,503]	73,649 [101–137,450]	170 [32–238]	3,764 [35–9,820]
Green	249	4,520 [37–104,981]	7,284 [36–150,692]	108 [17–400]	1,023 [21–8,621]
Orange	867	9,921 [37–64,706]	17,117 [55–129,067]	104 [14–571]	1,079 [19–9,530]
All	1198	7,942 [37–104,981]	13,713 [36–150,692]	108 [14–571]	1,070 [19–9,820]

Table 1: Descriptive statistics of the construction project schedules analysed here. The three companies are anonymised and identified by the colours used in Figures 3–6.

along the DUS spine; arcs above the spine represent extra dependencies between non-adjacent DUSs, with darker arcs encoding more dependencies. Because the DUS network has at most a few hundred nodes even for the largest schedules, it is small enough to draw, it offers which makes it a natural canvas on which to overlay other project information. In Figure 1C, D Figure 2 shows what we call the DUS profile of two real construction schedules: the DUS spine runs along the horizontal axis, the height of each bar is proportional to the number of activities in the corresponding DUS, and that level, the colour encodes the Physical Progress Percent, a standard progress metric in construction. We refer to this composite display as the progress, and the arcs above record dependencies that skip levels. Read this way, a 22DUS profile, 373-activity schedule becomes a single legible picture in which one can see where the work is concentrated, how far it has progressed, and where long-range coupling occurs.

Scaling properties

To probe how the DUS network scales with project size we collected 1198 schedules from three construction companies, identified below as blue, green and orange (see Methods). The number of DUSs grows only logarithmically Table 1 reports their descriptive statistics. The projects span 37 to 104,981 activities and 36 to 150,692 dependencies, close to four orders of magnitude,

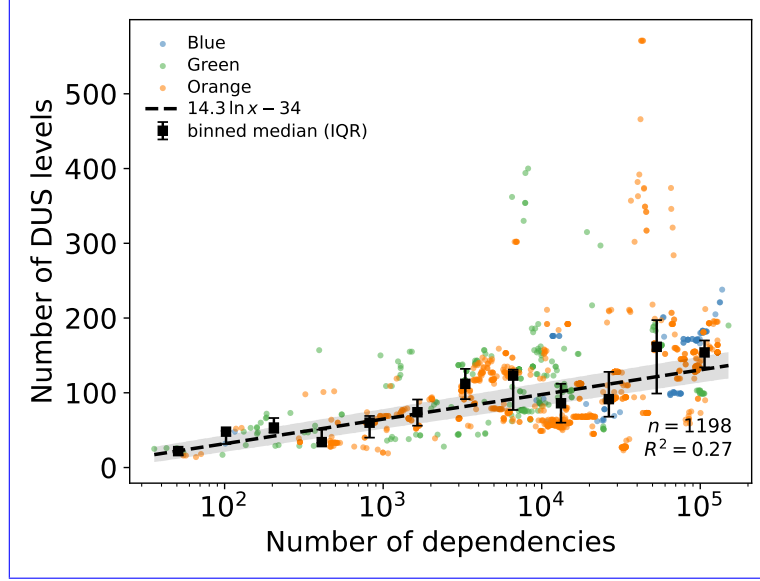


Figure 3: ~~Sealing of the~~ The number of DUSs and DUS levels grows logarithmically with the number of dependencies. Each point ~~represents a~~ is one of 1198 construction project projects, coloured by company. The dashed line is ~~a~~ the maximum-likelihood fit to $m = p \ln x + c$ under multiplicative errors, with $p = 14.3$ and $c = -33.9$; the ~~scaling law~~ $y = p \log(x)$ using all data points ~~grey band is the 95% bootstrap confidence region~~. Colours correspond to projects from three different construction companies ~~Black squares are binned medians with interquartile ranges~~.

and the three companies overlap substantially in size.

The number of DUS levels grows very slowly with the number of ~~activity-level~~ dependencies (Figure 3), ~~and the same relation describes all three companies~~. The DUS network is ~~therefore~~. Because a logarithm and a power law with a small exponent are hard to tell apart over a finite range, we compared four functional forms — a one-parameter logarithm, a two-parameter logarithm, a power law and a linear relation — each fitted by maximum likelihood under both additive and multiplicative errors, and selected between them by AIC (Methods). The two-parameter logarithm with multiplicative errors is preferred, with $m = 14.3 \ln x - 33.9$ where x is the number of dependencies. It is favoured over the power law by $\Delta\text{AIC} = 22$, over the one-parameter logarithm by $\Delta\text{AIC} = 45$, and over the linear relation by $\Delta\text{AIC} = 147$; its residuals show no detectable heteroscedasticity (Breusch–Pagan $p = 0.07$). The best-fitting power law has exponent 0.17, small enough that the practical distinction from a logarithm matters little over the observed range. The logarithmic coefficient is 14.3 with a 95% bootstrap confidence interval of [13.4, 15.2].

The relation is a statement about the central tendency, not about individual projects: it explains 27% of the variance in $\ln m$, and at a given number of dependencies the number of DUS levels varies by a factor of several. We therefore describe it as a slow, logarithmic growth of the central trend rather than as a scaling law in the strict sense. Even so, the compression it implies is substantial. Increasing the number of dependencies by a factor of ten adds about 33 DUS levels regardless of where one starts, so the DUS network is an exponential compression of the underlying dependency network: ~~doubling the size of the project adds only a constant number of DUS levels~~. Concretely, the median project in Table 1 has 13,713 dependencies and 108 DUS levels.

Fitting each company separately gives logarithmic coefficients of 15.9, 16.5 and 14.6. A likelihood ratio test of a model with a shared coefficient and company-specific offsets against a model with fully company-specific parameters does not reject the shared coefficient (LR = 2.4,

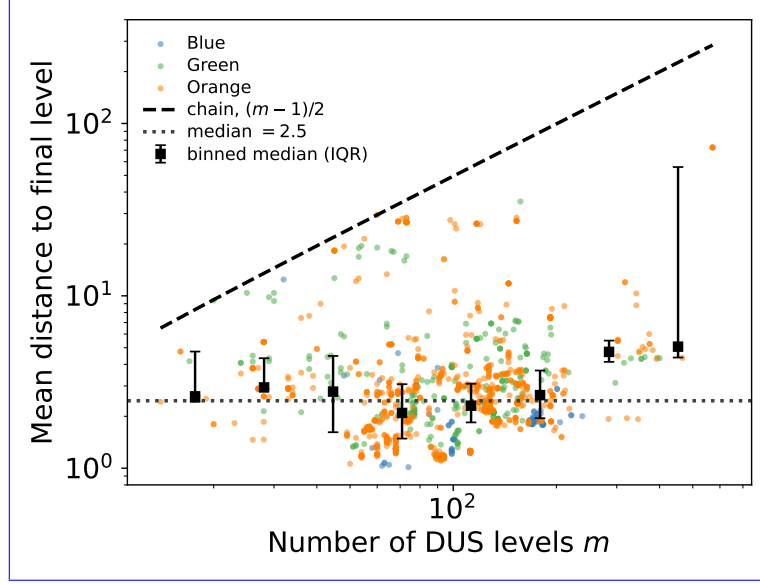


Figure 4: Dependencies that skip levels make the DUS networks are small-worlds network shallow. Average-Mean shortest distance between every from a DUS and level to the ending final DUS versus-level, against the number of DUSs in the project schedule DUS levels m . Each point represents The dashed line is the expectation $(m-1)/2$ for a construction project pure chain. Colours correspond to projects from three different construction companies The distance is essentially independent of m , with median 2.5 (dotted line). Black squares are binned medians with interquartile ranges.

$df = 2, p = 0.30$), and AIC prefers the shared-coefficient model. The offsets do differ significantly ($LR = 38.4, df = 2, p = 5 \times 10^{-9}$). So the rate at which levels accumulate with dependencies is common to the three companies at 15.1 per natural logarithm, while the absolute number of levels is shifted by company-specific scheduling practice.

A second question is how far apart, on average, two DUSs-DUS levels are in the network. If a project had no dependencies outside the DUS spine, the mean distance from any DUS shortest distance from a level to the final one would be exactly $(m-1)/2$. Real projects fall well far below this expectation (points vs dashed line in Figure 4): large DUS networks are much more tightly connected than their spines would suggest. The the median distance is 2.5 levels, with an interquartile range of [1.9, 3.6], and 99.8% of projects lie below the chain expectation, at a median of 4.6% of it. More striking than the gap is the near-absence of growth. Binned medians are 3.4 for projects with fewer than 50 levels, 2.3 for 50 to 150 levels, and 2.7 above 150 levels, even though m ranges from 14 to 571. The rank correlation between the two is small though nominally significant ($\rho = 0.10, p = 8 \times 10^{-4}, n = 1198$). Applying the model selection procedure used above, a weakly increasing linear relation is in fact preferred over a strictly constant one ($\Delta AIC = 20$), with slope 0.0041 per level (95% CI [0.0018, 0.0062]). The growth this implies is negligible in practice: across the entire observed range the fitted mean distance rises only from 2.5 to 4.7, a factor of 1.9, while the number of levels grows by a factor of 41.

The reason is that many dependencies act as shortcuts, linking DUSs-levels that are far apart along the spine (see Figure 1A, B). In this sense, the DUS networks of construction projects are small-worlds [28]—the arcs in Figure 2). This is the mechanism that produces short paths in small-world networks [28], and it answers a question a project manager might ask directly: a schedule with 500 DUS levels is not 500 steps deep in any practical sense, because a handful of long-range dependencies carry information from almost anywhere to the end in two or three steps.

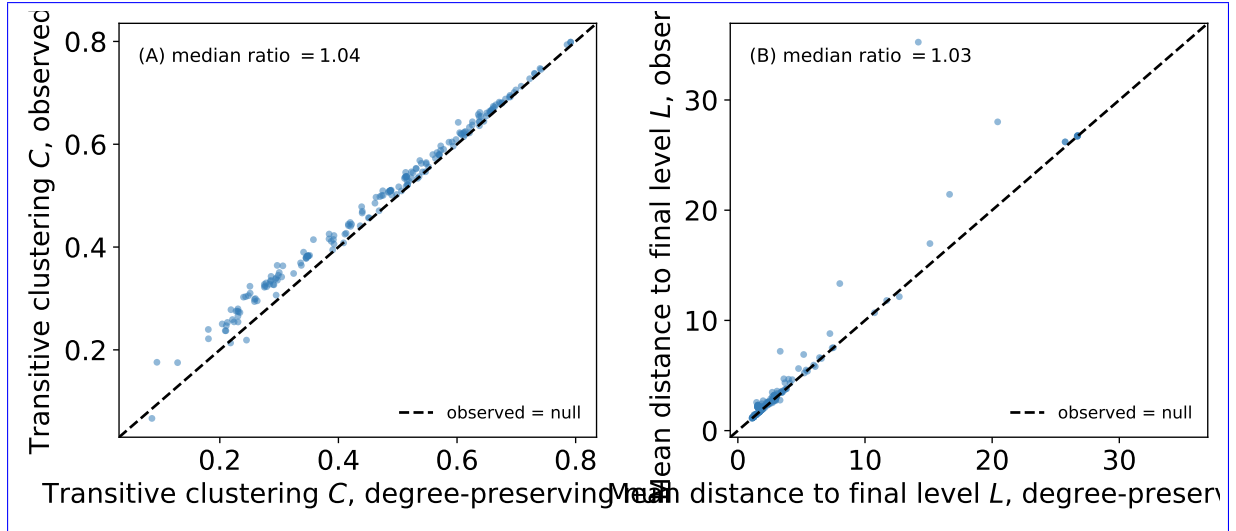


Figure 5: The shallowness follows from the degree distributions. Observed versus degree-preserving null values of (A) the transitive clustering coefficient C and (B) the mean distance to the final DUS level L , for 198 construction projects. The null holds the DUS spine fixed and rewires the skip dependencies while preserving every level’s skip in-degree and out-degree; each point is the average over 20 randomisations. Both quantities lie essentially on the identity line, so randomly reallocating the shortcuts reproduces what is observed.

What makes the DUS network shallow

Does this shallowness require the shortcuts to be arranged in some particular way? To find out we compared each DUS network with a degree-preserving null. Clustering in a DAG cannot be measured with the usual coefficient, because a cyclic triangle is impossible; a *transitive* triangle is not, since $i \rightarrow j$, $j \rightarrow k$ and the shortcut $i \rightarrow k$ are mutually consistent with acyclicity. We therefore use the transitive clustering coefficient C , the fraction of directed two-paths $i \rightarrow j \rightarrow k$ that are closed by an edge $i \rightarrow k$. It is the admissible member of the directed-triangle family, it is exactly the quantity a shortcut creates, and it is zero for a pure chain. The null holds the spine fixed, since Eq. (1) forces it and randomisation should not be free to destroy it, and rewires the skip dependencies by double-edge swaps that preserve every level’s skip in-degree and skip out-degree (see Methods).

Across 198 projects the observed values are almost exactly those of the null (Figure 5): $C/C_{\text{null}} = 1.04$ with an interquartile range of $[1.01, 1.10]$, and $L/L_{\text{null}} = 1.03$ with an interquartile range of $[1.00, 1.08]$. The short distances are therefore a consequence of the degree distributions of the DUS network rather than of any special placement of the shortcuts: given how many long-range dependencies each level has, putting them anywhere reproduces the observed distances. This closes a link in a mechanistic chain, because the duplication-split model of activity networks [29] generates degree distributions of just this kind, and those distributions are what the present result shows to be sufficient.

It also settles how the small-world question should be posed here. The Watts–Strogatz criterion asks for clustering substantially in excess of a degree-matched null, and that excess is absent, so the criterion is not the informative test for these networks and we state the property directly instead of invoking the label. The property itself is not in question: the mean distance to the end of the project is 2.5 levels where a chain of the same depth would give up to 285.

The degree sequence does not, however, account for everything. Under a stronger randomisation, which preserves the in-degree and out-degree of every *activity*, keeps the network acyclic and then re-derives the DUS decomposition from scratch, the resulting networks are roughly half as deep as the real ones (median $m/m_{\text{null}} \approx 2$, a conservative figure since the discrepancy widens

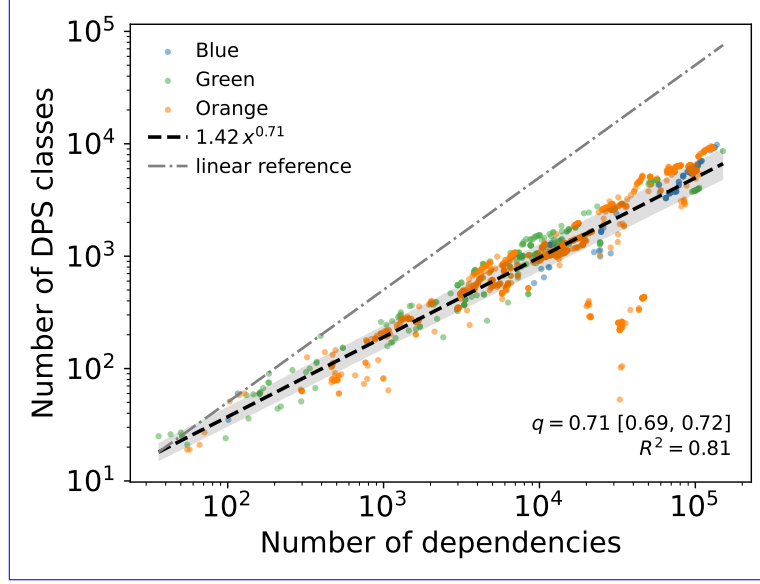


Figure 6: Nodes in The number of DPS classes grows as a dependency network are stratified based on their dependency patterns. Node colour indicates sub-linear power law with the DFS number of dependencies. Each point is one of 1198 construction projects, and coloured by company. The dashed line is the node label gives maximum-likelihood fit $1.42 x^{0.71}$ under multiplicative errors; the DUS grey band is the 95% bootstrap confidence region. The dash-dotted line is a linear reference of slope 1.

with further randomisation). Real schedules are thus considerably more sequential than a degree-matched random DAG. The degree distributions determine how short the paths through the DUS network are; they do not determine how deep it is.

DUS fine structure

Scaling of the number of DFSs and the number of dependencies. Each point represents a construction project. The dashed line is a fit to the scaling law $y = px^q$ using all data points. Colours correspond to projects from three different construction companies.

Activities sharing the same DUS index can be further split according to the split further according to which DUS levels their predecessors come from (Figure 41B). An activity at $DUS = i, u = i$ must have at least one predecessor at $DUS = i - 1, u = i - 1$; predecessors at $DUS = 0, \dots, i - 2, u = 0, \dots, i - 2$ are optional. Each of the $i - 1$ optional levels can be either present or absent, giving 2^{i-1} distinguishable predecessor patterns at the i -th DUS. (Activities at $DUS = 0$ level. Activities at $u = 0$ have no predecessors, and those at $DUS = 1, u = 1$ draw exclusively from $DUS = 0, u = 0$, so neither can be split further.)

These finer We call the resulting partition the *Dependency Predecessor Structure (DPS)* of the network, and each of its parts a DPS class. As with the DUS, we distinguish the DPS index of an activity, the DPS class as a set of activities, the DPS as the partition, and the DPS network defined below. These groupings are related to what mathematicians call *graph fibrations* [30, 31]; we refer to them as the *Dependency Fibration Structure (DFS)* of the network; two activities in the same DPS class have predecessor sets that occupy the same levels, a local symmetry of the kind that fibrations formalise and that has been used to find building blocks in biological networks. Restricting to DAGs again simplifies things simplifies matters, because

each ~~DFS~~ class can be labelled by a single integer. We set $f_i = 0$ when $u_i = 0$ and otherwise

$$f_i = \sum_{k=0}^{u_i-2} b_{ik} 2^k + 2^{u_i-1} \quad (2)$$

where $b_{ik} = 1$ if activity i has at least one predecessor in ~~the DUS level~~ level k ~~th DUS~~ and $b_{ik} = 0$ otherwise. The integer f_i should be read as a categorical label, not a metric: numerical proximity of two f_i values does not imply similarity of their predecessor patterns.

As before, we collapse the dependency network onto its ~~DFSs~~ DPS classes to obtain the ~~DPS~~ DPS network: each node is a ~~DPS~~ DPS class, and a weighted edge $F_a \xrightarrow{w_{ab}} F_b$ records the w_{ab} activity-level dependencies between them. Figure ~~5–6~~ plots the number of DPS classes in each project against the number of activity-level dependencies. The relation is sub-linear and well-described by a power law (dashed line) dependencies. The Applying the same model selection procedure as above, a power law under multiplicative errors is preferred, with $1.42 x^{0.71}$, ahead of the linear relation by $\Delta AIC = 269$ and the two-parameter logarithm by $\Delta AIC = 1310$. The exponent is 0.709 with a 95% bootstrap confidence interval of $[0.692, 0.724]$, comfortably below 1, and the fit explains 81% of the variance in the logarithm. The residuals of this fit are heteroscedastic, with more scatter at large sizes, so the confidence interval quoted here is a case-resampling bootstrap rather than a nominal standard error. As for the DUS, a shared exponent across the three companies is not rejected (LR = 1.1, df = 2, $p = 0.59$; shared exponent 0.72), while the prefactors differ ($p = 4 \times 10^{-6}$).

The growth is therefore sub-linear but much faster than logarithmic, placing the DPS network between the original dependency network and the much smaller DUS network. The theoretical bound 2^{i-1} on the number of patterns at level i is therefore loose in practice : the realised count of distinct DFSs grows much more slowly than the worst case, and most theoretically possible patterns by an enormous margin: a project with 100 DUS levels could in principle realise more than 10^{29} classes and instead realises of order 10^3 . Most predecessor patterns that are combinatorially possible simply do not occur in real schedules. The DFS network sits between the original dependency network and the much smaller DUS network in terms of size

Replication on public data

The construction schedules analysed here are proprietary. To make the central results reproducible from public data, we repeated the analysis on the 2040 single-mode instances of PSPLIB [19], the standard benchmark library for resource-constrained project scheduling, and on the two anonymised real schedules distributed with our code (Figure 7).

PSPLIB instances are much smaller than real construction schedules. They contain 32 to 122 activities and 48 to 257 dependencies, come in only four activity counts, and span barely half a decade in size; their dependency range overlaps only the extreme lower tail of ours, since just 2.4% of our construction projects (29 of 1198) are that small. Model selection within PSPLIB alone favours a power law for the number of DUS levels and a linear relation for the number of DPS classes, which are not the forms selected on the construction data. Over so narrow a range, and with only four distinct sizes, we do not read this as evidence about functional form. We use PSPLIB instead for the question it can answer, which is whether the construction relations transfer to independent data at a size where they were barely constrained.

For the DPS the answer is yes. The construction power law, evaluated two orders of magnitude below the median project on which it was estimated, predicts the PSPLIB values to a median observed/predicted ratio of 0.77 (Figure 7B), and the power law fitted within PSPLIB alone, though not the form AIC prefers there, has exponent 0.86, also sub-linear. For the DUS the answer is no: the construction logarithm over-predicts the PSPLIB values by more than a factor of two (median observed/predicted ratio 0.42, Figure 7A). The logarithmic relation should

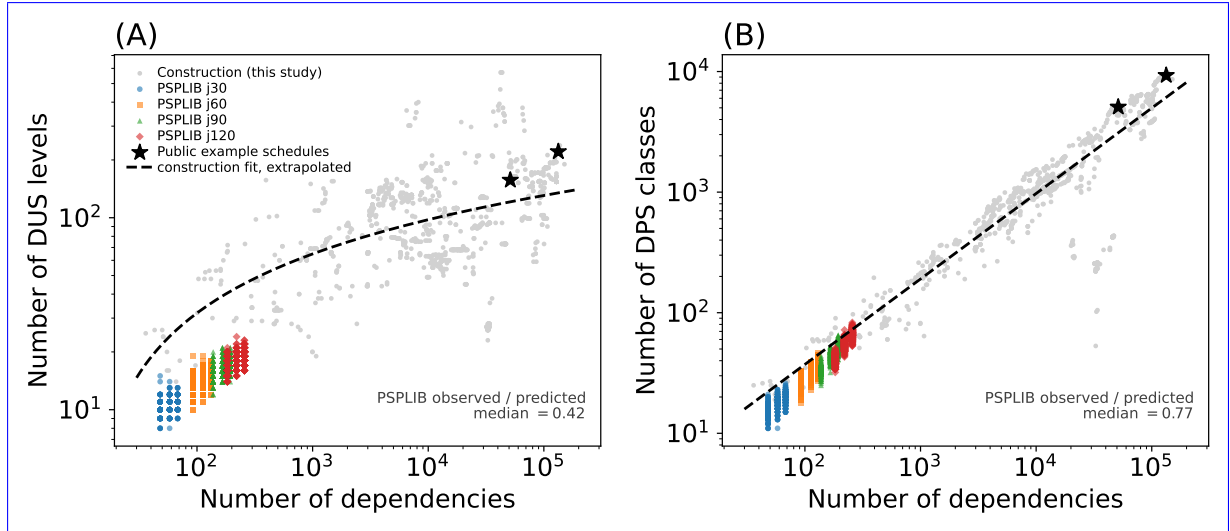


Figure 7: Replication on public benchmark instances. (A) Number of DUS levels and (B) number of DPS classes against the number of dependencies, for 2040 single-mode PSPLIB instances (coloured), the 1198 construction schedules of this study (grey), and the two anonymised real schedules distributed with the code repository (stars). Dashed lines are the fits to the construction data, evaluated two orders of magnitude below the median project size on which they were estimated. Only 2.4% of the construction projects are as small as the largest PSPLIB instance.

therefore be read as a description of large construction schedules rather than as a universal law, which is a further reason to avoid the term scaling law for it. The qualitative claims do transfer: PSPLIB DUS networks are also shallow, with a median distance to the final level of 2.1, below the chain expectation in 100% of instances.

Discussion

Dependency networks admit two natural higher-level representations forming a chain of progressively coarser views, Activity $\rightarrow$ DFS $\rightarrow$ DUS Activity $\rightarrow$ DPS $\rightarrow$ DUS. For construction project schedules, the DFS-DPS provides a sub-linear coarse-graining and the DUS an exponential coarse-graining of the underlying dependency network. Both can be used directly for tracking and reporting, and both can serve as inputs one.

The practical value of these constructions to project management is that they give a view of the schedule that the WBS cannot. A WBS reports progress by deliverable; the DUS reports it by logical depth, which is what determines what can start next. Three uses follow directly. First, reporting and tracking: the DUS profile of Figure 2 compresses a schedule of tens of thousands of activities into a picture with of order a hundred nodes, on which progress, cost or risk can be overlaid, and in which a stalled level or an unusually large concentration of work is immediately visible. Second, diagnosis: the arcs in Figure 2 are dependencies that jump many levels, and these are exactly the links whose delay propagates furthest; they are candidates for scrutiny because a long-range dependency is often either a genuine structural constraint or an artefact of how the schedule was built. Third, as an input to risk models that estimate project duration and cost., since a coarse-grained network with a hundred nodes is tractable for simulation where one with a hundred thousand is not. That the compression is deterministic and parameter-free matters here: two planners applying it to the same schedule obtain the same answer, which is not true of clustering methods that depend on an objective function or a random seed.

Three limitations should be kept in view. The relations reported here are established on schedules from three companies in one sector, and the failure of the logarithmic relation to extrapolate to PSPLIB shows that the coefficients are not universal. The DUS is one antichain decomposition among many, and we have not shown it is optimal by any criterion; it is justified by being canonical, cheap and interpretable rather than by an optimality argument. Finally, the null models used here randomise one structure at a time, holding either the DUS spine or the activity degree sequence fixed; they bound what those constraints alone imply, and are not a generative account of how real schedules come to have the degree distributions they do.

Because the construction is purely topological, the same machinery applies to any system ~~that can be represented~~ representable as a directed acyclic graph, including causal networks, ancestries and feedforward neural networks. ~~Two questions naturally follow. First, do the size scaling laws~~ Three questions follow. Do the size relations observed for construction schedules carry over to ~~these other settings?~~ Second, what those settings, and if the coefficients differ, what do the differences measure? What features of the underlying dependency network determine the structure of its ~~DFS~~ DPS and DUS networks, ~~including the scaling between them?~~? And since the degree distributions of the DUS network are enough to produce its short distances but not its depth, what generative mechanism fixes the depth? The duplication-split model [29] is a natural candidate, since its split operation creates sequential structure of exactly the sort the degree-preserving null destroys.

Methods

Dataset

The construction project schedules analysed here were drawn from the Nodes & Links database. Three construction companies are represented, identified throughout as blue, green and orange. ~~Each company contributed several projects, contributing 82, allowing variation to be assessed both within a single organisation and across organisations. 249 and 867 projects respectively, for a total of 1198. Table 1 reports the number of activities, dependencies, DUS levels and DPS classes for each company. One further schedule was excluded because its first DUS level connects directly to every other level, so that its DUS network is a star rather than a network with a spine.~~ Activity-level dependencies and Physical Progress Percent values were taken directly from the schedules as recorded in the database.

Computation of the DUS and DFS

The public replication uses the four single-mode PSPLIB instance sets j30, j60, j90 and j120 [19], comprising 2040 instances of 32, 62, 92 and 122 activities, downloaded from the PSPLIB distribution site. The precedence relations of each instance were extracted and treated exactly as the dependencies of a construction schedule.

Computation of the DUS and DPS

The DUS index u_i defined in ~~equation~~ Eq. (1) is computed by first sorting the activities into a topological order ~~using Kahn's algorithm~~, then sweeping through that order and applying Eq. (1) to each activity in turn. The cost is linear in the number of activities plus the number of dependencies. The ~~DFS~~ DPS index f_i defined in ~~equation~~ (2) Eq. (2) requires a single additional pass over the dependencies once u_i is known: for each activity i we record which lower DUS levels its predecessors occupy, and combine these bits into the integer label given by (2) Eq. (2). The DUS and ~~DFS~~ DPS networks are then obtained by collapsing the dependency network onto these labels and accumulating edge weights. Distances within the DUS network are shortest-path

distances in the unweighted directed network, computed by a single reverse sweep over the levels.

Code availability Null models for the DUS network

Two degree-preserving randomisations were used, both by double-edge swaps, which preserve in- and out-degree exactly.

The first randomises the DUS network itself. The spine $U_0 \rightarrow U_1 \rightarrow \dots$ is forced by Eq. (1) and is held fixed. The remaining skip dependencies are rewired by repeatedly choosing two of them, $a \rightarrow b$ and $c \rightarrow d$, and replacing them by $a \rightarrow d$ and $c \rightarrow b$ whenever both results are still forward skips and neither already exists. This preserves each level’s skip in-degree and skip out-degree and the acyclicity of the network. We used 20 swap attempts per skip edge and averaged over 20 randomisations per project; the resulting C and L are unchanged between 5 and 80 swap attempts per edge, so the chain is well mixed.

The second randomises the activity network. A topological order of the observed DAG is fixed, and pairs of dependencies are swapped whenever both resulting edges still point forward in that order, which preserves every activity’s in- and out-degree and guarantees the result is acyclic. The DUS decomposition is then recomputed from scratch. This randomisation mixes more slowly: the number of DUS levels is still drifting at 20 swap attempts per edge, and the gap between the observed and randomised networks widens as mixing proceeds, so the ratios reported for it are conservative bounds rather than converged estimates.

Clustering is measured by the transitive clustering coefficient, the number of directed two-paths $i \rightarrow j \rightarrow k$ closed by an edge $i \rightarrow k$ divided by the number of directed two-paths. Distances are shortest-path distances to the final DUS level, as elsewhere in the paper.

Statistical methods

For each relation we fitted four candidate functional forms: a one-parameter logarithm $y = p \ln x$, a two-parameter logarithm $y = p \ln x + c$, a power law $y = ax^b$, and a linear relation $y = ax + b$. Each was fitted by maximum likelihood under two error structures: additive, $y = f(x) + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, and multiplicative, $y = f(x)e^\varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$. The multiplicative log-likelihoods include the Jacobian term $-\sum_i \ln y_i$, so that all eight fits are likelihoods of the same observed quantity and their AIC and BIC values are directly comparable across both functional form and error structure. This addresses the fact that a naive ordinary least squares fit on log-log axes is not comparable, on its own, to a fit on the natural scale.

Confidence intervals are nonparametric case-resampling bootstraps with 2000 replicates, which do not assume homoscedastic residuals; this matters for the DPS relation, whose residuals are heteroscedastic. Heteroscedasticity was assessed with a Breusch–Pagan test of the squared residuals against $\ln x$.

Commonality of parameters across companies was assessed with nested likelihood ratio tests under the selected error structure, comparing three models: a pooled model with one shared parameter pair; a model with a shared logarithmic coefficient, or power-law exponent, and company-specific offsets or prefactors; and a model with fully company-specific parameters. This separates the question of whether the companies share the form of the relation from the question of whether they share its nuisance parameters. We report the likelihood ratio statistic, its degrees of freedom and its p value, together with the AIC of each model.

We did not apply the Clauset–Shalizi–Newman procedure [18], which is designed to fit and test heavy-tailed probability distributions of a single variable. The relations studied here are regressions of one measured quantity on another across projects, not distributions, so that procedure does not apply; the analogous concerns it raises about naive log-log fitting are addressed by the explicit error-model comparison and bootstrap intervals described above.

Code and data availability

A Python implementation of the DUS and ~~DFS~~ DPS algorithms, together with ~~helper code to draw fish plots~~ the two anonymised construction schedules used in Figure 2 and helper code for the DUS profile plots, is available at <https://github.com/av2atgh/dus>. The scripts that perform the statistical analysis, download and process the PSPLIB benchmark, and generate every figure in this paper are included in the same repository. The PSPLIB instances are publicly available from <https://www.om-db.wi.tum.de/psplib/>. The construction project schedules are proprietary to Nodes & Links Ltd and its clients and cannot be released; Table 1 reports their summary statistics, and Figure 7 shows that the main results are reproducible on public data.

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Author contributions

A.V. conceived the study, designed and implemented the algorithms, performed the analysis, prepared the figures and wrote the manuscript.

Use of generative AI

The author used Claude (Anthropic) for language editing of the manuscript text and for assistance with the statistical analysis code and figure generation scripts. The tool was not used for the conception of the study ~~or the design of the algorithms, the analysis of the data, the generation of figures, or the interpretation of results. All~~. All analyses were verified by the author, and all scientific content, claims and conclusions are the responsibility of the author.

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